

Day 2: Game Theory

Main points of this session:

Game Theory is the formal study of strategic interaction.

There are five elements of a game: Players, Strategies, Information/beliefs, Payoffs, Rationality.

Strategy A *dominates* strategy B for player N iff for every action of other players, N is better off playing A than B.

Nash equilibrium - game outcome when no player wants to deviate given all other players keep their actions.

Learning objectives for today:

- Define a game and be able to explain what Game Theory studies;
- Consider several topics and settings Game Theory studies;
- Demonstrate Game Theory role in real life and its applications;
- Discuss the idea and be able to explain the fact that there can be a better outcome for everyone than the outcome the society ends up with if everyone optimizes.

Game Theory is the formal study of strategic interaction.

There are five elements of a game:

1. **Players:** who is interacting? How many of them are here?
2. **Strategies:** what are the options for each player?
3. **Payoffs:** where can the game end? How does each player like each game end?
4. **Information/beliefs:** what do players know at each decision point? What do they assume about the situation, each other, etc?
5. **Rationality:** how do players think? ...do they think at all? :)

Nash equilibrium outcome is the outcome when no player wants to deviate given all other players keep their actions.

Strategy A **dominates** strategy B for player N iff for every action of other players, N is better off playing A than B. In Nash Equilibrium, every player plays a dominant strategy.

Individual optimization does not equal society's optimal:

A socially desirable outcome does not necessarily equal the equilibrium one. A famous example: Prisoners' Dilemma (advertising, etc).

There's a lot more to Game Theory:

Trigger strategies, mixed strategies, behavioral economics explaining beliefs and rationality, limited information games, and many more things!